

This assignment will not be graded and is only for practice.

Note: Suppose W_1 and W_2 are subspaces of a vector space V . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that $W_1 + W_2$ is a subspace of V .

Multiple Select Questions (MSQ):

1) Let $V_1 = \mathbb{R}^3$ be a vector space, define a function $\|\cdot\|_i : V_1 \rightarrow \mathbb{R}$ for $i = 0, 1, 2, 3$ as: **1 point**

$$\begin{aligned}\|(x, y, z)\|_0 &= x + y + z \\ \|(x, y, z)\|_1 &= |x| + |y| + |z| \\ \|(x, y, z)\|_2 &= \sqrt{x^2 + y^2 + z^2} \\ \|(x, y, z)\|_3 &= \max\{|x|, |y|, |z|\}\end{aligned}$$

and let $V_2 = M_{2 \times 2}(\mathbb{R})$ be a vector space, define a function $\|\cdot\|_i : V_2 \rightarrow \mathbb{R}$ for $i = 4, 5$ as:

$$\begin{aligned}\|A\|_4 &= \max\{|a_{11}| + |a_{12}|, |a_{21}| + |a_{22}|\}, \text{ where } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}. \\ \|A\|_5 &= \max\{|a_{11}| + |a_{21}|, |a_{12}| + |a_{22}|\}, \text{ where } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.\end{aligned}$$

Which of the following option(s) is (are) true.

- ☐ $\|\cdot\|_0$ is a norm on V_1 .
- ☐ $\|\cdot\|_1$ is not a norm on V_1 .
- ☐ $\|\cdot\|_2$ is a norm on V_1 .
- ☐ $\|\cdot\|_3$ is not a norm on V_1 .
- ☐ $\|\cdot\|_4$ is not a norm on V_2 .
- ☐ $\|\cdot\|_5$ is a norm on V_2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\|\cdot\|_2$ is a norm on V_1 .

$\|\cdot\|_5$ is a norm on V_2

2) Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{R}^3$, i.e., $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_1 + x_2y_2 + x_3y_3$. Consider a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined as: **1 point**

$$T(u) = \langle u, v \rangle,$$

where $v \in \mathbb{R}^3$. Which of the following options is (are) true for T ?

- ☐ T is one-one for all $v = 0 \in \mathbb{R}^3$.
- ☐ T is onto for all $v = 0 \in \mathbb{R}^3$.
- ☐ T is not one-one for every $v \in \mathbb{R}^3$.
- ☐ There exists a $v \in \mathbb{R}^3$ such that T is an isomorphism.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is onto for all $v = 0 \in \mathbb{R}^3$.

T is not one-one for every $v \in \mathbb{R}^3$.

3) Which of the following options are true?

1 point

$V = \mathbb{R}^2$ and a function defined as:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

$$\langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^2 x_i y_j.$$

is not an inner product on the vector space V .

- ☐ $V = M_{2 \times 2}(\mathbb{R})$ and $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products on the vector space V , a function defined as: $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_1 + \langle \cdot, \cdot \rangle_2$ is an inner product on the vector space V .
- ☐ $V = M_{3 \times 3}(\mathbb{R})$ and $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products on the vector space V , a function defined as: $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_1 - \langle \cdot, \cdot \rangle_2$ is an inner product on the vector space V .
- ☐

$V = \mathbb{R}^2$ and a function defined as:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

$$\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 - x_2 y_1.$$

is an inner product on the vector space V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \mathbb{R}^2$ and a function defined as:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

$$\langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^2 x_i y_j.$$

is not an inner product on the vector space V .

$V = M_{2 \times 2}(\mathbb{R})$ and $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products on the vector space V , a function defined as: $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_1 + \langle \cdot, \cdot \rangle_2$ is an inner product on the vector space V .

4) Consider a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as $T(x, y, z) = (x + 2y, x - y + cz, 2x + y + dz)$. Let A be the matrix representation of T with respect to the standard

1 point

ordered bases of $\mathbb{R}^3$ and $B = \begin{bmatrix} 0 & 3 & 2 \\ -1 & -3 & -3 \\ 2 & 3 & 4 \end{bmatrix}$.

Based on the above information which of the followings option(s) is(are) true?

- ☐ If A is similar to B , then rank of the linear transformation T is 2.
- ☐ If A is similar to B , then nullity of the linear transformation T is 2.
- ☐ If $c = d$, then A cannot be similar to B .
- ☐ If A is similar to B and $c = 1$, then a basis of the kernel is $\left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If A is similar to B , then rank of the linear transformation T is 2.

If $c = d$, then A cannot be similar to B .

If A is similar to B and $c = 1$, then a basis of the kernel is $\left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right\}$.

5) Consider a vector $v = (1, 2)$ from the inner product space $\mathbb{R}^2$, where inner product is the dot product. Which of the following options is/are true? **1 point**

- ☐ There exists a vector $u (= v)$ in $\mathbb{R}^2$ such that $||u|| = ||v||$.
- ☐ There are infinitely many vectors in $\mathbb{R}^2$ such that norms of the vectors are equal to norm of v .
- ☐ There exists a vector u in $\mathbb{R}^2$ such that $\langle u, v \rangle = 0$.
- ☐ There are infinitely many vectors in $\mathbb{R}^2$ such that angle between the vector and v is equal to 90° .

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a vector $u (= v)$ in $\mathbb{R}^2$ such that $||u|| = ||v||$.

There are infinitely many vectors in $\mathbb{R}^2$ such that norms of the vectors are equal to norm of v .

There exists a vector u in $\mathbb{R}^2$ such that $\langle u, v \rangle = 0$.

There are infinitely many vectors in $\mathbb{R}^2$ such that angle between the vector and v is equal to 90° .

6) Let L and L' be affine subspaces of $\mathbb{R}^3$, where $L = U$ and $L' = (2, 0, 1) + U'$, for some vector subspaces U and U' of $\mathbb{R}^3$. Let a basis for U be given by $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ and a basis for U' be given by $\{(1, 0, 1), (0, 1, 1)\}$. Suppose there is a linear transformation $T : U \rightarrow U'$ such that $(0, 1, 0) \in \ker(T)$, $T(2, 0, 1) = (0, 1, 1)$ and $T(1, 1, 0) = (1, 0, 1)$. An affine mapping $f : L \rightarrow L'$ is obtained by defining $f(u) = (2, 0, 1) + T(u)$, for all $u \in U$. Which of the following options are true? **1 point**

- ☐ $L = \{(x, y, z) \mid x - y - 2z = 0\}$
- ☐ $L = \mathbb{R}^3$
- ☐ $L' = \{(x, y, z) \mid x + y - z = 1\}$
- ☐ $L' = \mathbb{R}^3$
- ☐ $f(x, y, z) = (x - 2z + 2, z, x - z + 1)$

☐ $f(x, y, z) = (x - 2z + 2, \frac{x}{2}, x - z + 1)$

☐ $f(x, y, z) = (x - 2z, z, x - z)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$L = \mathbb{R}^3$$

$$L' = \{(x, y, z) \mid x + y - z = 1\}$$

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$$

7) Let $L_1 = \{(x, y, z) \mid x + y + 2z = 6\}$ and $L_2 = \{(x, y, z) \mid 2x + y + 2z = 9\}$ be affine subspaces of $\mathbb{R}^3$ and W_1 and W_2 be the corresponding vector subspaces of the given affine subspaces of L_1 and L_2 respectively. Choose the set of correct options **1 point**

☐ $W_1 = \{(x, y, z) \mid x + y + z = 0\}$

☐ $W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$

☐ $W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$

☐ $W_2 = \{(x, y, z) \mid 2x + y + 2z = 1\}$

☐ $W_1 = \{(x, y, z) \mid x + y + 2z = 1\}$

☐ $W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$$

$$W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$$

$$W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$$

Numerical Answer Type (NAT):

8)

Let θ be the angle between the vectors $u = (2, 1, -3)$ and $v = (5, 8, 4)$ and $||\cdot||$ denote the usual length in $\mathbb{R}^3$ with respect to the dot product, then what is $||u|| ||v|| \cos(\theta)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

9) Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{R}^3$, i.e., $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3$. Let $v_1 = (1, 0, -1)$, $v_2 = (a, b, c) \in \mathbb{R}^3$. Suppose the length of the vector v_2 is 3 and $\langle v_2, (1, -1, 1) \rangle = -3$, $\langle v_2, (-1, 2, 1) \rangle = 1$. If the angle between v_1 and v_2 is 45° , then find the value of $a + b + c$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

10) Let $A_{4 \times 4}$, $B_{4 \times 4}$ and $C_{4 \times 4}$ be matrices such that A and B are equivalent matrices and B and C are similar matrices. If rank of the matrix A is 3 then find the determinant of the matrix C .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Your score is: 0/10